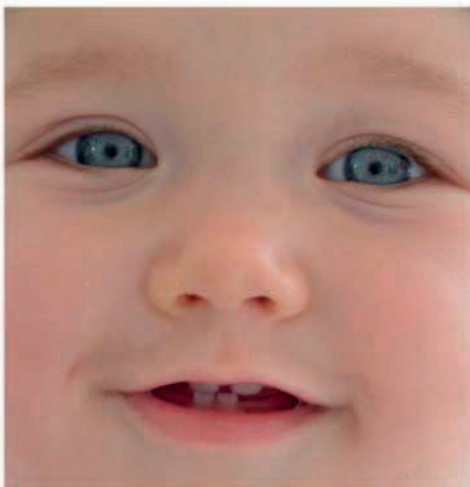


Q What color will my baby's eyes be and what determines this?

You baby's eye color depends on how dominant and recessive alleles (variants of genes—see p.166) work.

Each person has two alleles for eye color (and other characteristics), one inherited from each of their parents. The brown allele (B) is dominant and the blue allele (b) is recessive, so when the two come together (Bb) brown dominates over blue and the result is brown eyes. In order to have blue eyes, a person must have two blue alleles (bb). Therefore, if your baby has blue eyes but both parents have brown eyes, both you and your partner must be Bb—each of you has one brown allele and one blue allele.

A Bb mother and a Bb father have a 25-percent chance of having a blue-eyed baby. It becomes a 50-percent chance if one of you is bb and the other Bb. If one or both of you is BB, the only possible outcome is a baby with brown eyes. Of course, if you both have blue eyes (you are both bb), your baby will have blue eyes, too.



Blue eyes Many Caucasian babies' eyes look blue or gray when they are first born. It can take up to a year for the pigmentation to settle in and reveal the true eye color.

Blue and green are both recessive alleles. When blue and green are the only alleles present, they are actually codominant, so your baby's eye color could take on blue, green, or a mix of the two.

EYE COLOR

The diagram below illustrates how genes determine your baby's eye color. This principle

PARENTS' EYE COLOR AND GENE

BB  + BB 

bb  + bb 

Bb  + bb 

Bb  + BB 

Bb  + Bb 

Q Why is it that not all brown eyes are the same shade of brown?

No one is really sure. While the overall color of your baby's eyes (brown, hazel, blue, or green) might be determined by one set of genes, the level of pigmentation in your baby's iris may be determined by another—perhaps many others. Pigmentation comes from melanin, which is present not only in the iris of the eye, but in hair and skin, too. So, if you have two babies and the first has dark brown eyes and the second has light brown eyes, both babies have

dominant brown alleles in the eye-color gene. However, certain other genes caused different amounts of melanin in the eyes of the two children—resulting in different shades of brown. The more melanin that's released, the darker the pigmentation in the eye.

Q Will our baby inherit his father's height? Can we tell at birth?

Both parents' genes have a role to play in a person's final height as an adult. The mid-parental height formula estimates a child's potential growth—you simply add both parents' height in inches together, divide by two, and then add an extra 2½ inches for a boy or subtract 2½ inches for a girl. For instance, if mom's height is 5'4" and dad's is 5'11", a son's potential height would be 5'10" and a daughter's would be 5'5". This formula only gives a rough estimate, however, and in reality your child's health, diet, and other environmental factors play a part in addition to genes. Your baby's

size at birth doesn't necessarily indicate what his adult height will be, either, especially if he is premature. Mom's health, weight, and diet in pregnancy are more likely to have an affect on your baby's birth weight than genes.

Q Could my baby have curly hair if mine is straight?

In Caucasians, curly hair is determined by a dominant allele, so the answer to this depends on what pair of alleles your baby inherits from both parents. If you have straight hair, but your partner has curly hair, your baby is 50 to 75 percent likely to have curly hair, depending upon whether your partner has two curly-hair alleles, or one curly and one straight. If you both have straight hair, then your baby's hair will probably be straight, too. If you both have curly hair, the chances are that your baby will have curly hair—although if you both are straight-hair carriers, there's still a 25-percent chance that your baby's hair will be straight.

Color blindness is a condition that is passed down from parent to child and is much more common in men than women.